Assignment No: 1

Name: Atharv Shinde
Class: T.Y IT-A
Batch: D
Roll No: 33076
Subject: DSML

Code:

```
import pandas as pd
import numpy as np
import matplotlib.pyplot as plt
import seaborn as sns
df = pd.read_csv(r"C:\Users\admin\Documents\Titanic.csv")
print("First 5 Records:")
print(df.head())
```

O/p:

```
First 5 Records:
   PassengerId  Survived  Pclass  \
0            1         0       3
1            2         1       1
2            3         1       3
3            4         1       1
4            5         0       3

                                                Name     Sex   Age  SibSp  \
0                            Braund, Mr. Owen Harris    male  22.0      1
1  Cumings, Mrs. John Bradley (Florence Briggs Th...  female  38.0      1
2                             Heikkinen, Miss. Laina  female  26.0      0
3       Futrelle, Mrs. Jacques Heath (Lily May Peel)  female  35.0      1
4                           Allen, Mr. William Henry    male  35.0      0

   Parch            Ticket     Fare
0      0         A/5 21171   7.2500
1      0          PC 17599  71.2833
2      0  STON/O2. 3101282   7.9250
3      0            113803  53.1000
4      0            373450   8.0500
```

Code:

```
print("\nDataset Information:")
print(df.info())
```

O/p:

```
...
    Dataset Information:
    <class 'pandas.core.frame.DataFrame'>
    RangeIndex: 891 entries, 0 to 890
    Data columns (total 10 columns):
     #   Column       Non-Null Count  Dtype
    ---  ------       --------------  -----
     0   PassengerId  891 non-null    int64
     1   Survived     891 non-null    int64
     2   Pclass       891 non-null    int64
     3   Name         891 non-null    object
     4   Sex          891 non-null    object
     5   Age          714 non-null    float64
     6   SibSp        891 non-null    int64
     7   Parch        891 non-null    int64
     8   Ticket       891 non-null    object
     9   Fare         891 non-null    float64
    dtypes: float64(2), int64(5), object(3)
    memory usage: 69.7+ KB
    None
```

Code:

```
Q1 = df['Fare'].quantile(0.25)
Q3 = df['Fare'].quantile(0.75)
IQR = Q3 - Q1

df = df[(df['Fare'] >= Q1 - 1.5*IQR) & (df['Fare'] <= Q3 + 1.5*IQR)]

print("Dataset after removing outliers:")
print(df.shape)
```

O/p:

Dataset after removing outliers:
(775, 10)

Code:

```
bins = [0, 12, 19, 35, 60, 100]
labels = ['Child', 'Teen', 'Young Adult', 'Adult', 'Senior']
```

```
df['AgeGroup'] = pd.cut(df['Age'], bins=bins, labels=labels)

print(df[['Age', 'AgeGroup']].head())
```

O/p:

```
...      Age     AgeGroup
    0  22.0  Young Adult
    2  26.0  Young Adult
    3  35.0  Young Adult
    4  35.0  Young Adult
    5   NaN          NaN
```

Code:

```
print("\nAdditional Summary Statistics")

for col in ['Age', 'Fare']:
    print(f"\nColumn: {col}")
    print("Mean:", df[col].mean())
    print("Median:", df[col].median())
    print("Standard Deviation:", df[col].std())
    print("Variance:", df[col].var())
    print("Range:", df[col].max() - df[col].min())
```

O/p:

Additional Summary Statistics

Column: Age
Mean: 28.946574225122347
Median: 28.0
Standard Deviation: 14.368139363458516
Variance: 206.44342876776608
Range: 79.58

Column: Fare
Mean: 17.82209070967742
Median: 13.0
Standard Deviation: 13.57808535378775
Variance: 184.36440187474545
Range: 65.0

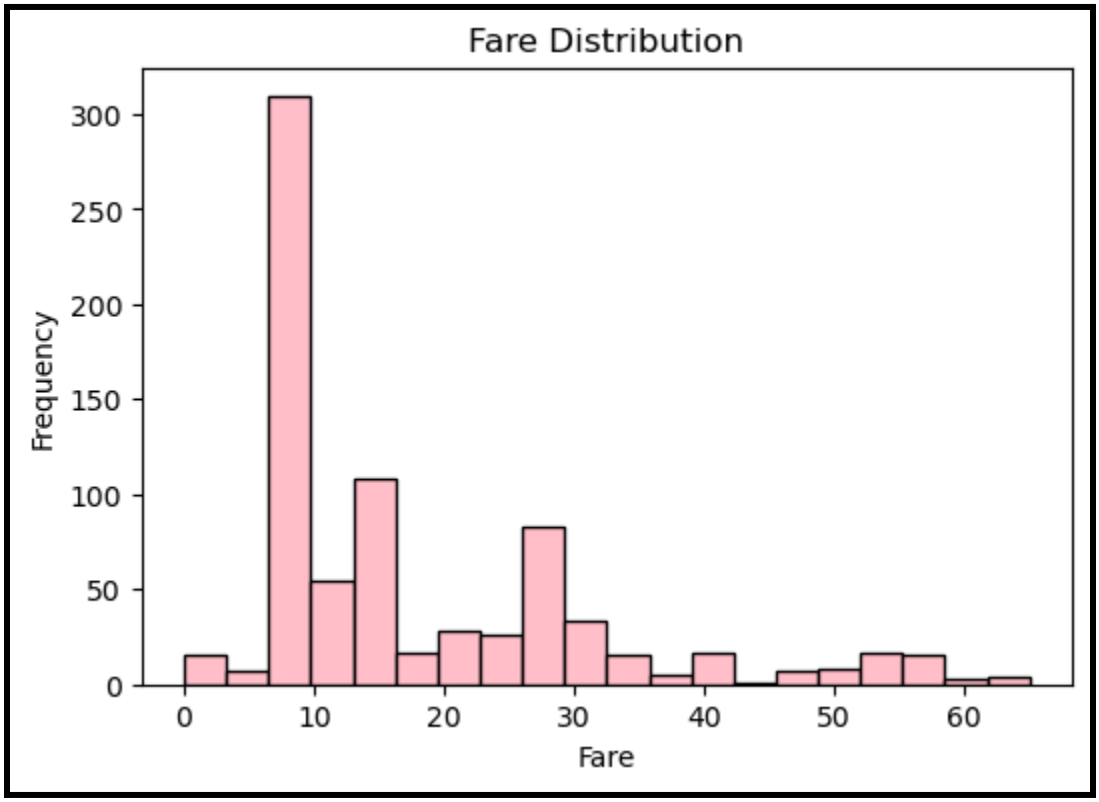
Code:

```
plt.figure(figsize=(6,4))
plt.hist(df['Fare'], bins=20, color='pink', edgecolor='black')

plt.title("Fare Distribution")
plt.xlabel("Fare")
plt.ylabel("Frequency")

plt.show()
```

O/p:



Code:

```
pclass = df['Pclass'].value_counts()

plt.figure(figsize=(5,5))

plt.pie(
    pclass,
```

```
    labels=['Class 3', 'Class 1', 'Class 2'],
    autopct='%1.1f%%',
    startangle=90
)

plt.title("Passenger Class Distribution")

plt.show()
```

O/p:

